

The Interrogation — Session 4 Lab Sheet

Saturday 5 September · your Part C output IS your P3, committed before 5:00 PM

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Ground rules: every chart you make must be *read* — a Markdown sentence under it saying what it shows and what it means for prediction. This session forward, **reading cells are graded**. Every predictor gets the calendar test: *knowable on the morning you predict?* Datasets load from Drive, never get committed. Nothing changes silently — print your counts.

PART A · GUIDED — YOUR SLICE, ON THE TABLE

Interrogate your own table (2:30–3:15)

Work on YOUR P2 slice. *No P2 slice yet? Build it now with the quick-start below, then continue — your P2 package is still owed separately.*

```
# QUICK-START (only if you have no slice yet): branch-month revenue
import pandas as pd
URL = "https://drive.google.com/uc?export=download&id="
sales = pd.read_csv(URL + "1LlA5ni-10NzZq60DU4eq8AlB6zi0WlDA")
sales = sales.drop_duplicates()
sales["BranchID"] = (sales["BranchID"].str.strip()
                    .str.upper().str.replace("-", ""))
sales["Date"] = pd.to_datetime(sales["Date"], errors="coerce")
sales = sales.dropna(subset=["Date"])
sales["YM"] = sales["Date"].dt.to_period("M").astype(str)
df = sales.groupby(["BranchID", "YM", as_index=False])["Amount"].sum()
```

A1 Reload & re-handshake · 5 min

1. Load your slice, run `info()` and your row count. Rows: **108** Does it match your P2? If not, stop and find out why before anything else.

Yes. The reloaded slice has 108 branch-month rows, exactly matching the validated P2 output.

A2 Univariate — is the average honest? · 12 min

```
df["<measure>"].describe()
df["<measure>"].skew()
df["<measure>"].hist(bins=30)
```

2. Mean: **97.72** Median: **94.50** Skewness: **0.10**

3. One sentence: does the mean fairly represent a typical row of your slice — and what would a model trained to “predict the average” get wrong?

The mean fairly represents a typical branch-month because it is close to the median and the skewness is only 0.10, but a model that only predicts the average would miss the large differences across branches and months, where orders range from 34 to 182.

A3 The rhythm — what repeats? · 13 min

Plot your measure over time (line per branch/product, or monthly bars). Hunt for: trend up/down · seasonality · a break in the pattern.

4. The strongest repeating pattern in your slice, WITH a number attached (e.g., “December runs ~25% above the monthly average”):

The strongest repeating pattern is the December increase, with an average of 129 orders per branch, about 32.0% above the overall monthly average of 97.72.

A4 The outlier hunt — dirt or loud truth? · 15 min

```
q1, q3 = df["<measure>"].quantile([.25, .75])
fence_hi = q3 + 1.5*(q3 - q1)
df[df["<measure>"] > fence_hi]
```

5. Rows beyond the fence: **1**. Pick your most extreme row and rule on it — KEEP / FLAG / DROP — with the reason a reader would accept:

Most extreme row: B01 · December 2025 · 182 orders

Decision: KEEP — December already shows a repeating seasonal increase, so 182 orders is consistent with real demand rather than an obvious data error.

Remember the demo: the fence is a flashlight, not a judge. An outlier that is real is signal about how big days happen — deleting it teaches your model that they don't.

PART B · HUNT THE LEAK

Ten columns walk in. Five are lying. (3:25–3:50)

The scenario: it is the FIRST MORNING of the month. The Operations lead must predict this month's revenue per branch (MonthRevenue_TARGET) to set staffing and stock. The file `hunt_the_leak.csv` (FILE_ID posted in the LMS) offers ten candidate predictor columns. **Five of them cheat.**

Your weapon is ONE question, applied to every column: ***“Could Kape Tayo actually know this number on the morning of day 1?”*** Correlation may rank candidates — the calendar disqualifies them. (One planted leak has an innocent, near-zero correlation. Correlation will not save you.)

6. Complete the audit — verdict and one-line reason for every column:

Column	Knowable day 1?	Verdict	Why (one line)
MonthNo	Yes	KEEP	The calendar month is already known.
PromoDaysPlanned	Yes	KEEP	Planned promotion days are scheduled in advance.
StaffHoursScheduled	Yes	KEEP	Scheduled staff hours are set before the month begins.
PrevMonthOrders	Yes	KEEP	The previous month's order count is already complete.
PrevMonthRevenue	Yes	KEEP	The previous month's revenue is already available.
OrderCount (this month)	No	LEAK	This month's orders are only known as the month happens.
AvgTicket (this month)	No	LEAK	It requires transactions from the month being predicted.
MonthSalesTotal	No	LEAK	It is the month's sales total and therefore contains the outcome being predicted.
RefundsProcessed	No	LEAK	Refunds happen during or after the month's transactions.
QuarterRevenueTotal	No	LEAK	The quarter total can include revenue from the target month and later months in the quarter.

7. Name the SNEAKIEST leak you found — the one most likely to survive into a careless model — and say what makes it sneaky:

QuarterRevenueTotal is the sneakiest leak because it looks like a normal historical summary, but its time window can already include the month being predicted. A beginner could keep it without noticing that part of the information would not exist on day 1.

PART C · INDIVIDUAL — YOUR P3

The EDA package (3:50–4:40)

P3 = an EDA notebook on YOUR slice with four deliverables. The Discussion → Code → Reading rhythm is the graded format — a notebook of bare code caps at Developing regardless of what the code does.

C1 Three annotated charts (minimum) · 20 min

Each chart earns its place: titled, axes labeled with units, and a **reading cell under it** answering two questions — what does it show, and what does it mean for PREDICTION? Suggested trio: your A2 distribution · your A3 rhythm · one comparison of your choosing (branch vs branch, product vs product, weekday vs weekend).

C2 Leakage audit — on your own predictors · 10 min

List every candidate predictor your P1 field 7 named (plus any new ones EDA suggested), and run the calendar test on each — same table as Part B, in your notebook:

My candidate predictor	Knowable in advance?	Verdict	Why
Historical inventory level	Yes	KEEP	Past inventory records exist before the prediction month.
Historical staffing level	Yes	KEEP	Past staffing records exist before the prediction month.
Historical storage condition	Yes	KEEP	Historical storage-condition records already exist.
BranchID	Yes	KEEP	The branch being forecast is known beforehand.
MonthNo	Yes	KEEP	The calendar month is known and can represent seasonality.
PrevMonthOrders	Yes	KEEP	The previous month's orders are complete before the new month starts.

C3 Base rate & balance note · 8 min

Compute YOUR track's base rate and write one sentence on what it means for your future model:

Track	Your base rate to compute	Why it matters
P4 · MKT	Weekend share of total revenue (%)	The naive "weekends matter" floor
P5 · FIN	Revenue share of the top 2 products (%)	Concentration — a 12-product forecast may really be a 2-product forecast
P6 · HR	Smallest branch headcount share (%)	Small groups = unstable rates — the imbalance lesson at branch grain
P7 · OPS	Quietest ÷ busiest branch-month order ratio	The spread a single flat forecast would smear over

8. My base rate: 0.187 (18.7%) Implication, one sentence:

The quietest branch-month has only 18.7% as many orders as the busiest, so a single flat forecast would hide a very large amount of demand variation across branches and months.

C4 The predictive-signal memo · 10 min

Five sentences, in your notebook, addressed to your P1 reader — fill each slot with a NUMBER, not an adjective:

Slot	Your sentence
1 · Strongest signal found	"The clearest predictable pattern is ... (magnitude: ...)"
2 · Second signal	"There is also ... (magnitude: ...)"
3 · One trap found & disqualified	"I considered ... but it fails the calendar test because ..."

Slot	Your sentence
4 • The floor to beat	"Any model must beat ... (my baseline / base rate: ...)"
5 • Modeling hypothesis	"Week 6, I will first try predicting ... using ..."

C5 Package and commit • 7 min + the commit block

P3 checklist — in <your-folder>/p3-eda/: ☐ notebook (Restart & Run all clean, reading cells throughout) ☐ ≥ 3 annotated charts ☐ leakage audit table ☐ base-rate note ☐ signal memo ☐ AI-use log entry WITH conversation link. Commit: "P3: <surname> – EDA, leakage audit, signal memo v1". Content freezes 5:00 PM; copy-editing until 11:59 tonight.

Next Saturday is the PRELIM (Rubric A): your P2 + P3, defended in a 10-minute oral — every decision, without notes. The work you commit today is the work you will stand behind next week. Build it like the panel is already listening.